

DEREK WANG

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Technical Skills

Languages: Java, C++, Python, HTML5, CSS, JavaScript

Developer Tools: Github, VS Code

Libraries/Frameworks: ReactJS, OpenCV

Engineering Tools: Fusion 360, MATLAB

Work Experience

City of Richmond Hill

March 2024 – Current

Youth Badminton Instructor

Rouge Woods Community Center, Richvale Community Center

- Coach a class of **14 beginner to intermediate students** (ages **9–16**) in **8-week terms**, focusing on footwork, technique, and strategy
- Designed a **new badminton curriculum** centered around **real game situations**, which was adopted by **all** badminton classes at Rouge Woods and Richvale, enhancing student engagement
- Provide personal **feedback and mentorship** while delivering lessons to improve **individual performance**, fostering an inclusive space

Extra Curricular Experiences

Checkmate Robotics Club | C++ | Github

September 2024 – Current

- Software team member of Team **16868C Rushdown** for the **VEX Robotics Competition**, invited to the **2025 VRC World Championships**
- Utilized **Okapi PROS Lib** to program robot functionalities, **Odometry** to track robot positioning, **PID Controllers** for precise motion control and error handling
- Wrote and tuned 15-second autonomous routes for matches, and 60-second programming skills, ranking **5th out of 80 teams** at **Provincial Championships**

Dr. X Robotics Academy | Java

July 2019 – May 2022

- Hardware team member of Team **16417 Dr. X Academy** for the **FIRST Tech Challenge**, invited to the **2022 FTC World Championships**
- Created **8 iterations** of the team's robot, including versions of the **drivetrain, arm, claw, turret** and a **backup robot**
- Write and tuned 30-second autonomous period for matches, using **java**, implementing **Dead Reckoning** for robot localization, and **PID Controllers** for motion control

Bayview Robotics Club | Instagram | Github

September 2021 – Current

- Led robotics team as **Vice President and main coder** in various robotics competitions such as **FIRST Tech Challenge** and **York Region Skills Challenge**
- Mentor **15 general members** and organize weekly STEM workshops, teaching them robotics basics in hardware and software.

Bayview Computer Club | Instagram

September 2024 – Current

- Plan and Deliver bi-weekly lessons on competitive programming **data structures and algorithms** to **20 members**, preparing them for the Canadian Computer Competition and future coding interviews

Projects

WebTyper | JavaScript | HTML | Source Code

- Utilized **Javascript, CSS, and HTML** to turn any website into an interactive typing game
- Keeps track of past statistics such as **Words Per Minute(WPM), Accuracy, and total times played**

Education

University of Watreloo

Waterloo, ON, Canada

Bachelor of Applied Science in Computer Engineering

Expected Graduation: April 2030

- Relevant Coursework: